

LAB 4 – VIBE CODING

Submission Report

Word Scramble Arena – Pygame

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Lab	Lab 4 – Vibe Coding
Project	Word Scramble Arena

1. Submission Links

Before Video Link:

<https://drive.google.com/file/d/1queOwyE5aN8qLcmGTnfXde1P1gOPE1gh/view?usp=drivesdk>

After Video Link:

https://drive.google.com/file/d/1VI85Dqg4kTgfMTENZGZy_tumC4jYTKRo/view?usp=drivesdk

Prompt Evidence / Public Chat History: <https://chatgpt.com/share/6ac64edf-6a24-83ee-8a44-88bd677a3e4e>

GitHub Repository: <https://github.com/dhonimahi/SE-Labs-PES1UG25CS808>

Lab 4 Code: <https://github.com/dhonimahi/SE-Labs-PES1UG25CS808/tree/main/Lab4>

2. Features / Bugs Fixed

Task 1 – Guess Validation Bug

Fixed the validation logic so the submitted answer is compared with the actual secret word rather than the scrambled word. Correct unscrambled answers are accepted.

Task 2 – Progressive Hint System

Added a HINT button. A hint reveals one letter in its correct position and applies a 0.25-point score penalty. The feedback now explicitly identifies the revealed letter and its position.

Task 3 – Countdown Timer

Added a 30-second round countdown. The remaining time is displayed during play. When the timer reaches zero, the correct word is revealed and the next round starts after a short delay.

Task 4 – Interactive Letter Tiles

Replaced the plain scrambled-word display with graphical letter tiles. The player can select one tile and another tile to swap their positions before submitting the answer.

3. Features / Bug-Fixed Code Snippets

Task 1 – Correct validation

```
is_correct = (guess == self.secret_word)
```

Task 2 – Clear hint feedback

```
revealed_letter = self.secret_word[position]

self.feedback_msg = (
    f"Hint: Letter '{revealed_letter}' is in "
    f"position {position + 1}! "
    f"-{self.hint_penalty:.2f} points."
)
```

Task 3 – Timeout handling

```
if self.time_left <= 0:
    self.time_left = 0
    self.round_expired = True
    self.feedback_msg = (
        f"Time's up! The word was "
        f"'{self.secret_word}'."
    )
```

Task 4 – Tile swapping

```
(
    self.tile_letters[self.selected_tile],
    self.tile_letters[index]
) = (
    self.tile_letters[index],
    self.tile_letters[self.selected_tile]
)
```

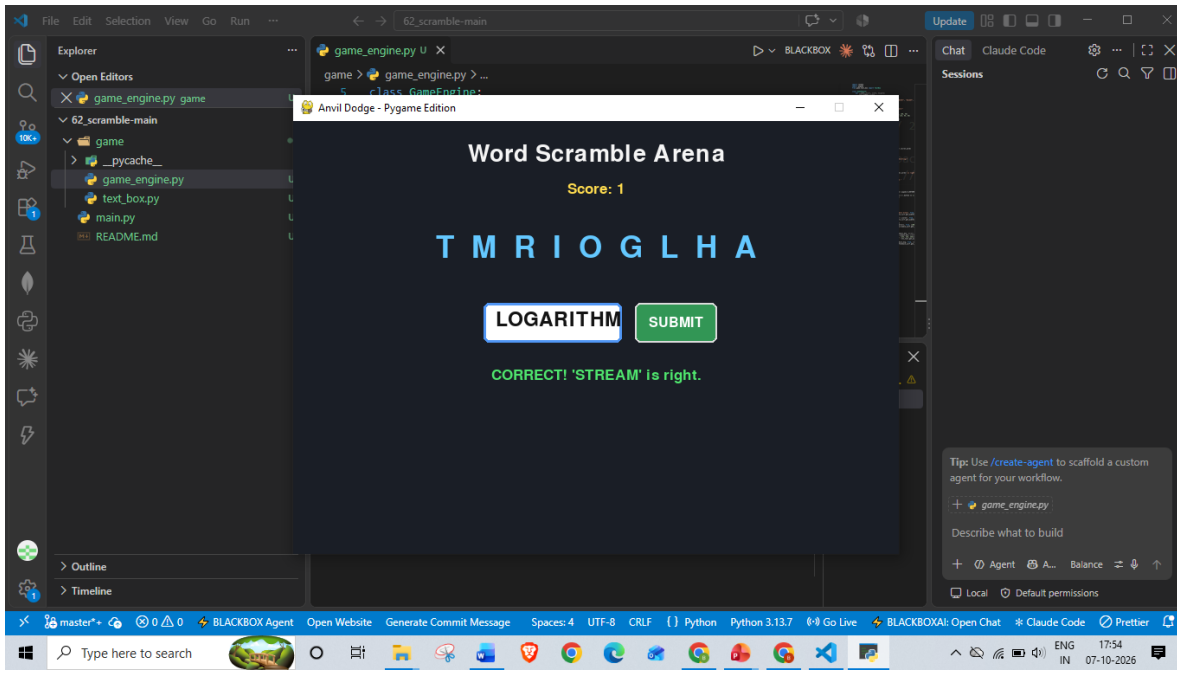
4. Testing Evidence

Task 1: Correct and incorrect guess behavior was tested.

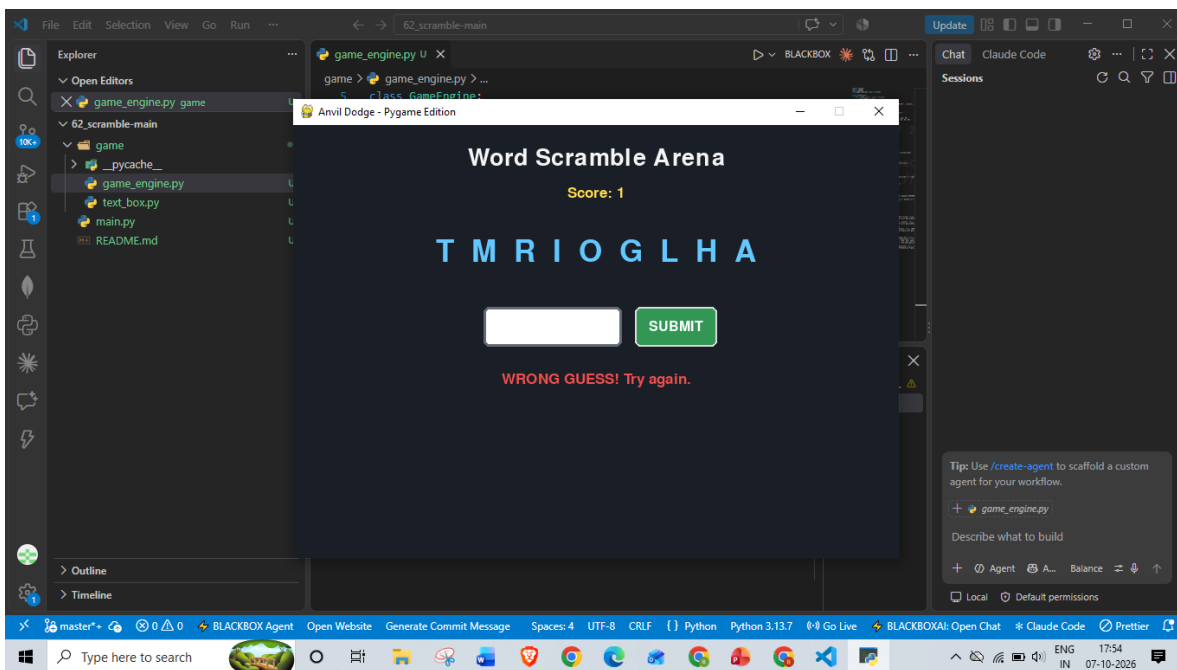
Task 2: Hint usage, revealed letters/positions, and the 0.25 score penalty were tested.

Task 3: The countdown was observed during play and timeout behavior was tested at 0.0 seconds.

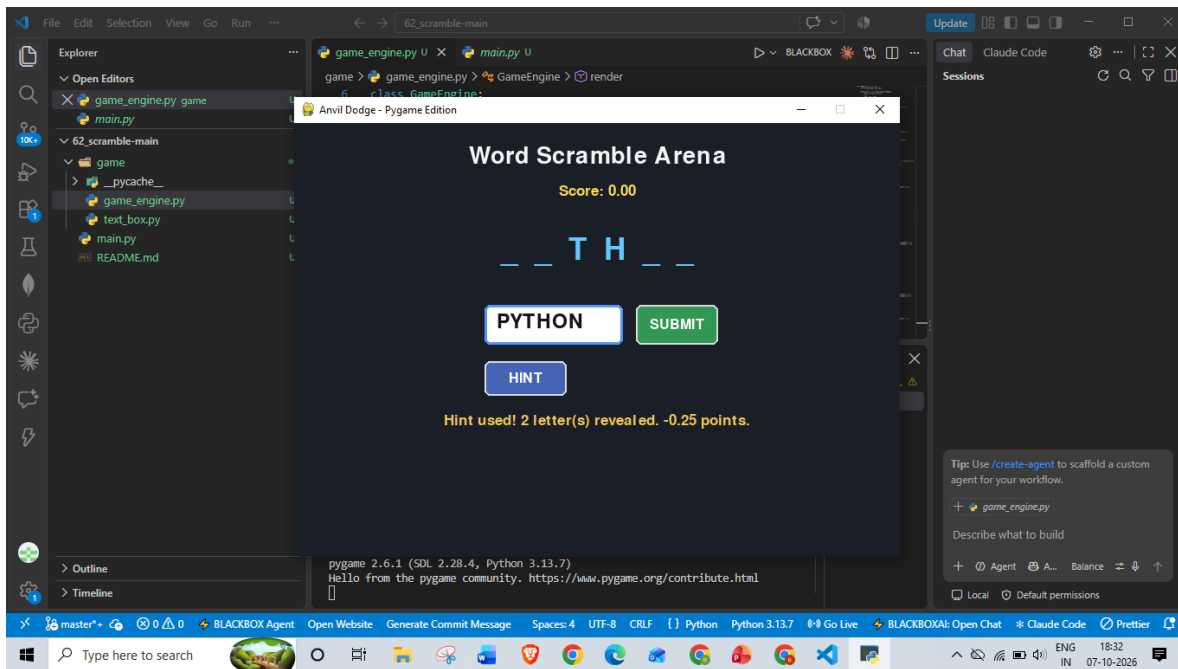
Task 4: Individual letter tiles were displayed and used for interactive rearrangement.



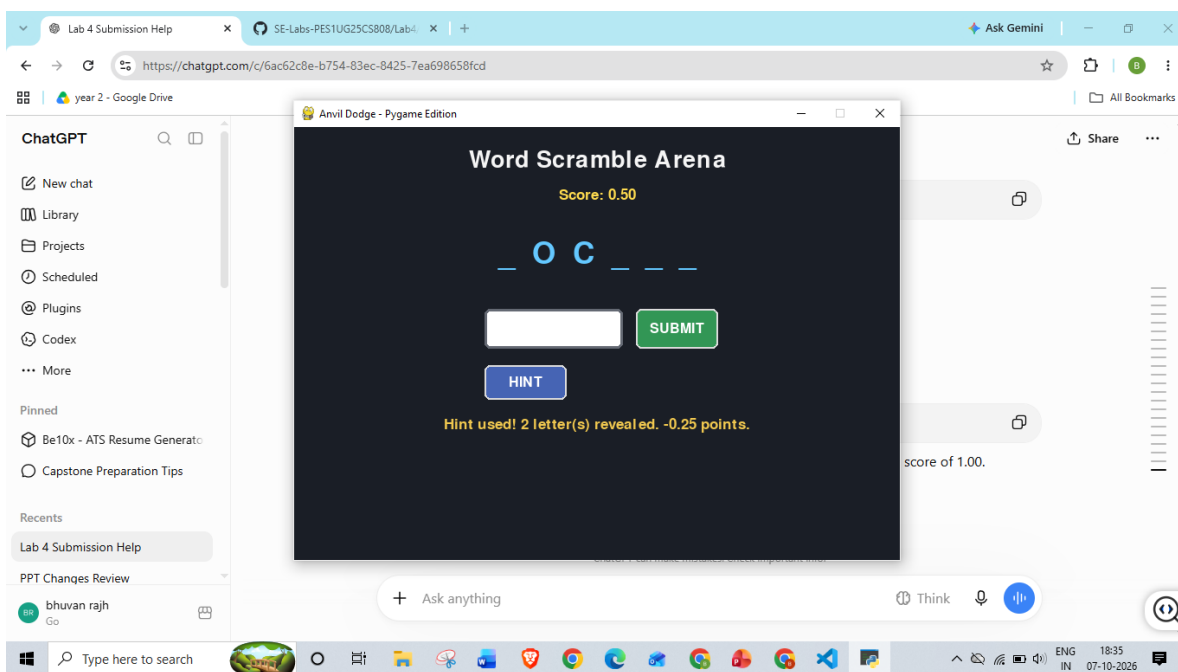
Task 1 – Correct guess accepted and score increased.



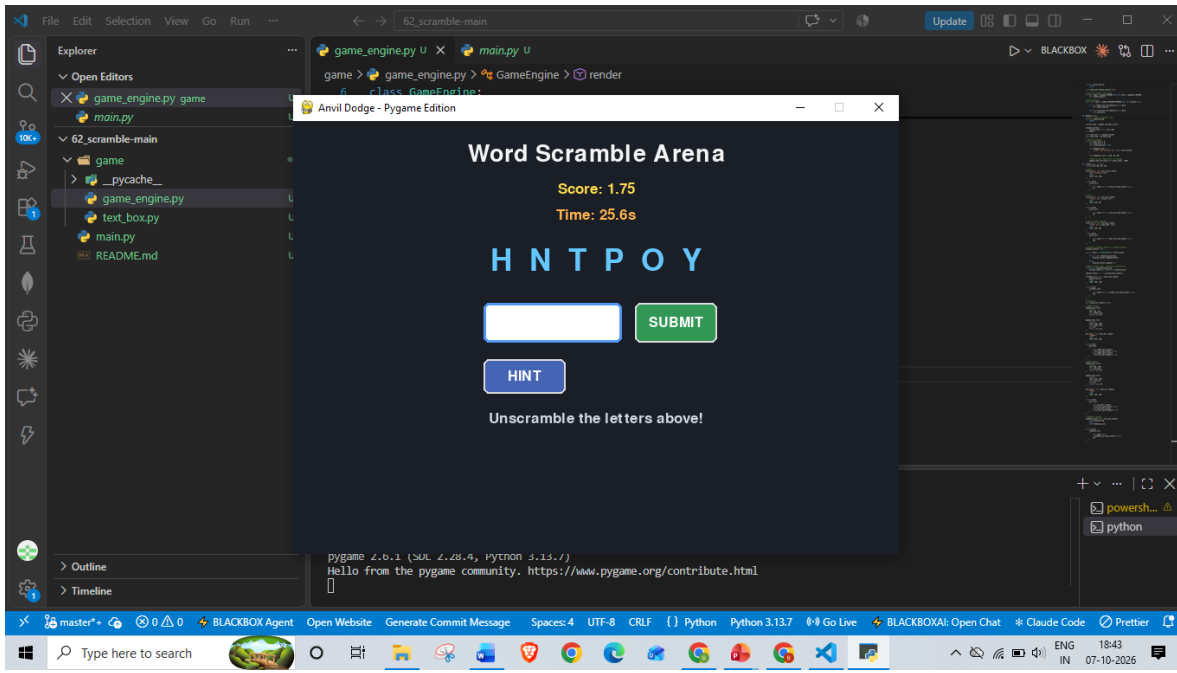
Task 1 – Incorrect guess feedback shown.



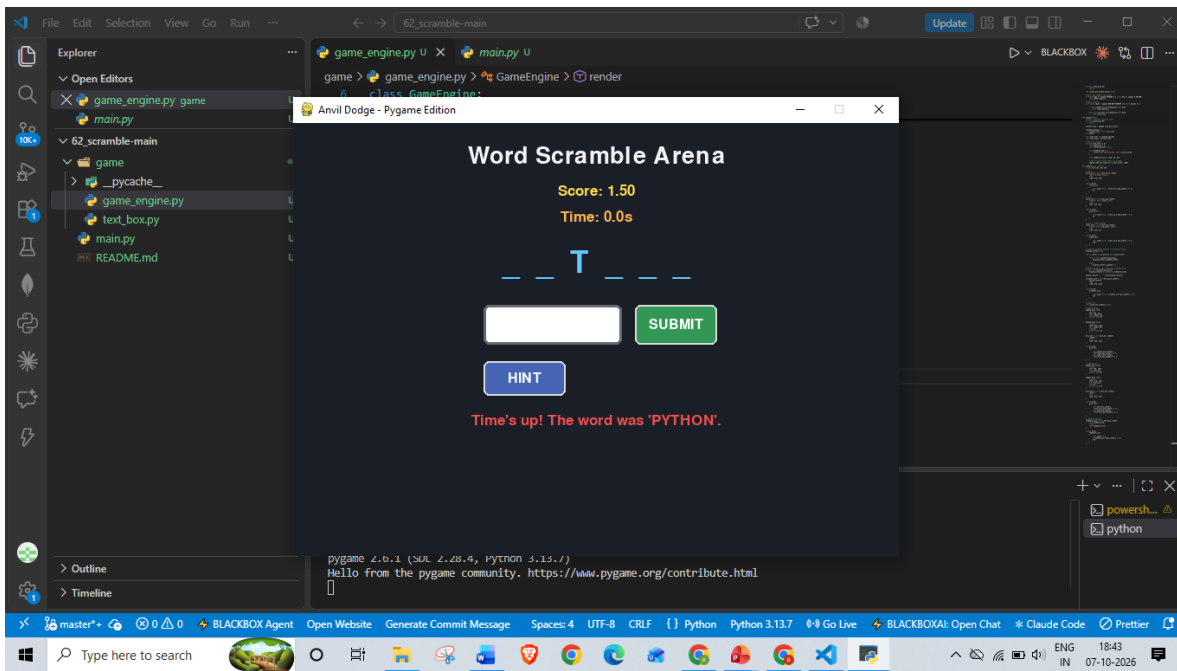
Task 2 – Hint feature showing revealed letters and score penalty.



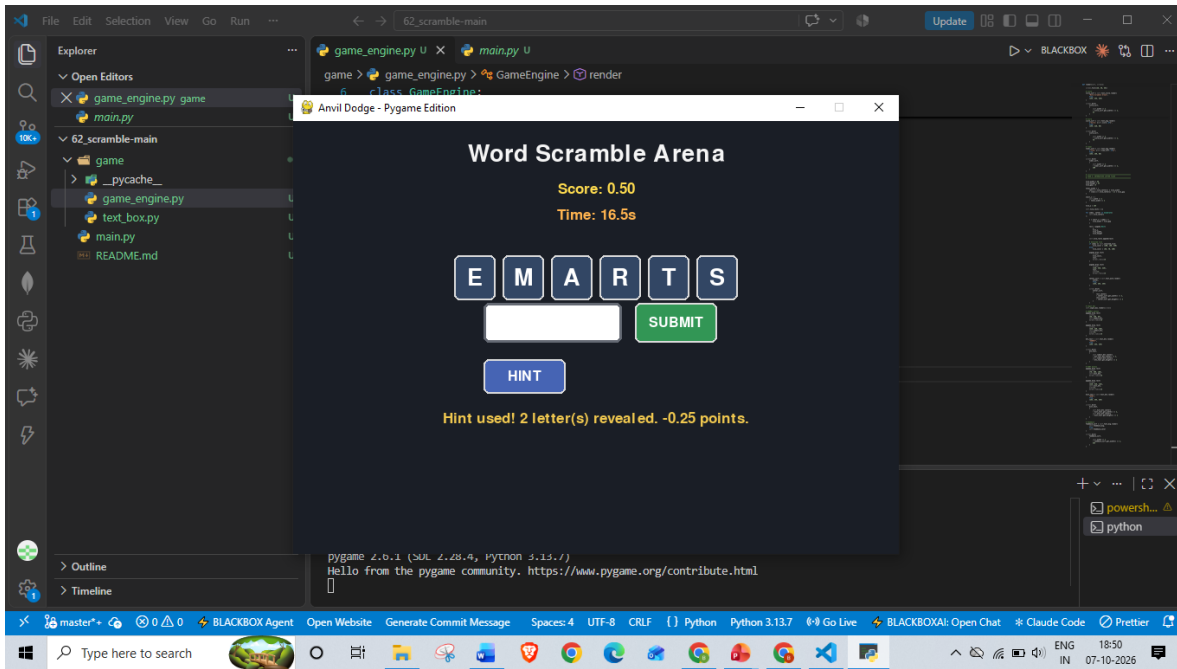
Task 2 – Hint behavior visible in the running application.



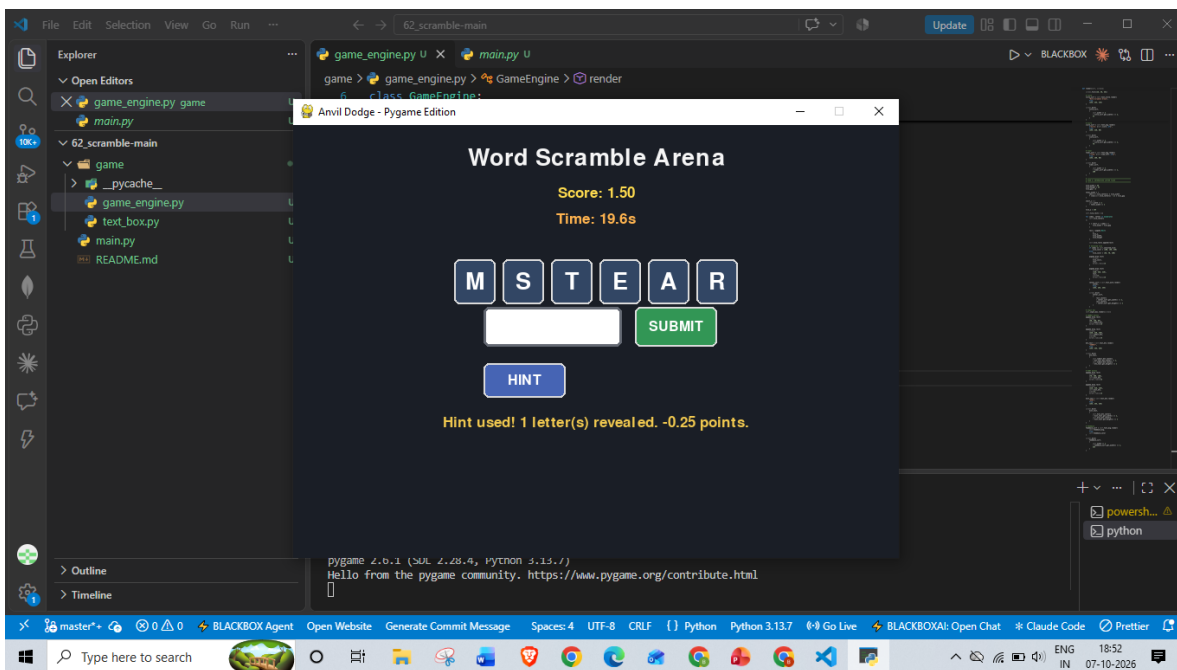
Task 3 – Countdown timer visible during a round.



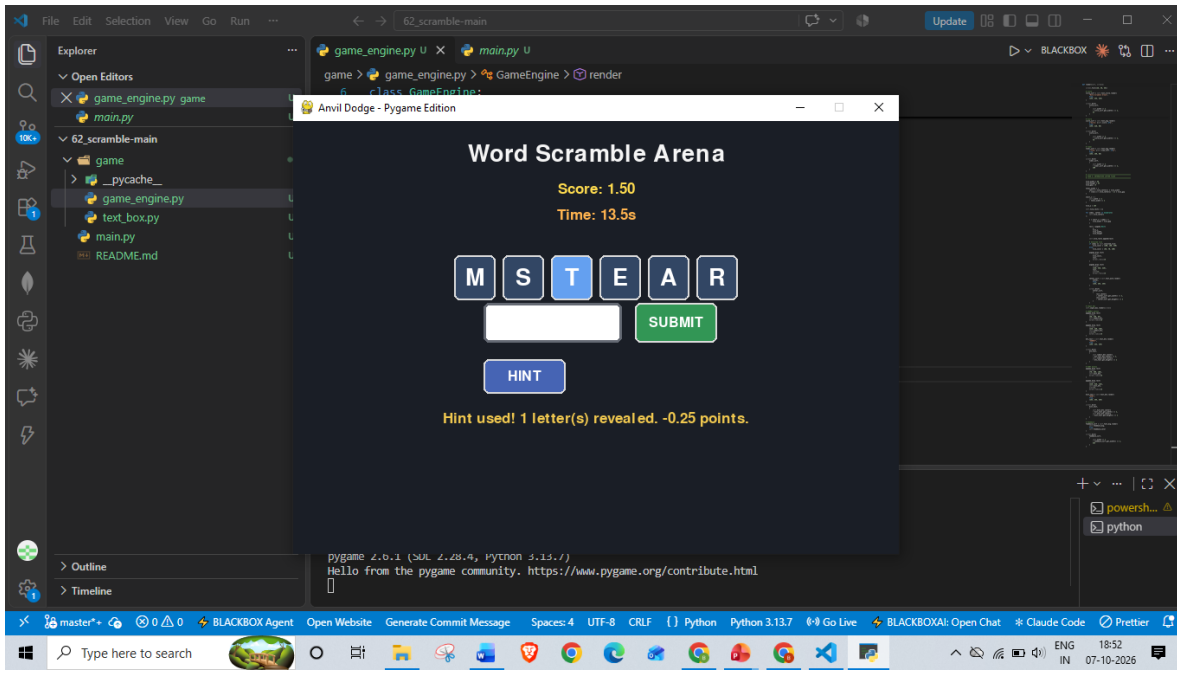
Task 3 – Timer reached 0.0 seconds and the correct word was revealed.



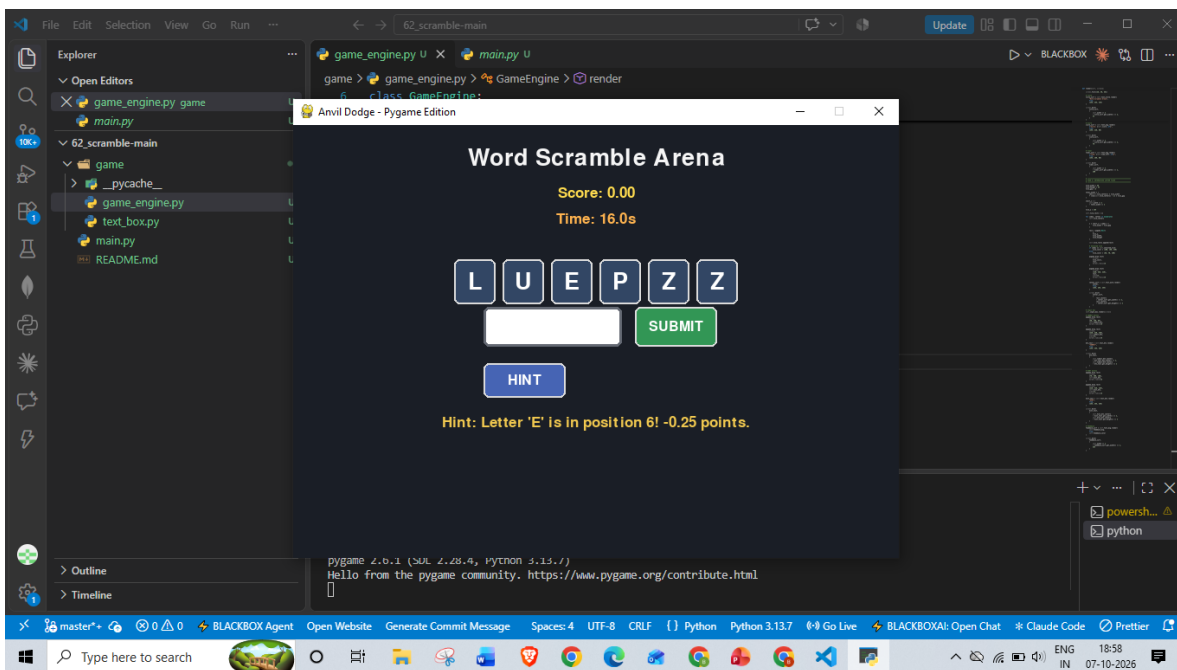
Task 4 – Graphical letter tiles are displayed individually.



Task 4 – Letter tiles arranged interactively.



Task 4 – Selected tile highlighted while rearranging letters.



Hint feedback explicitly identifies the revealed letter and position.

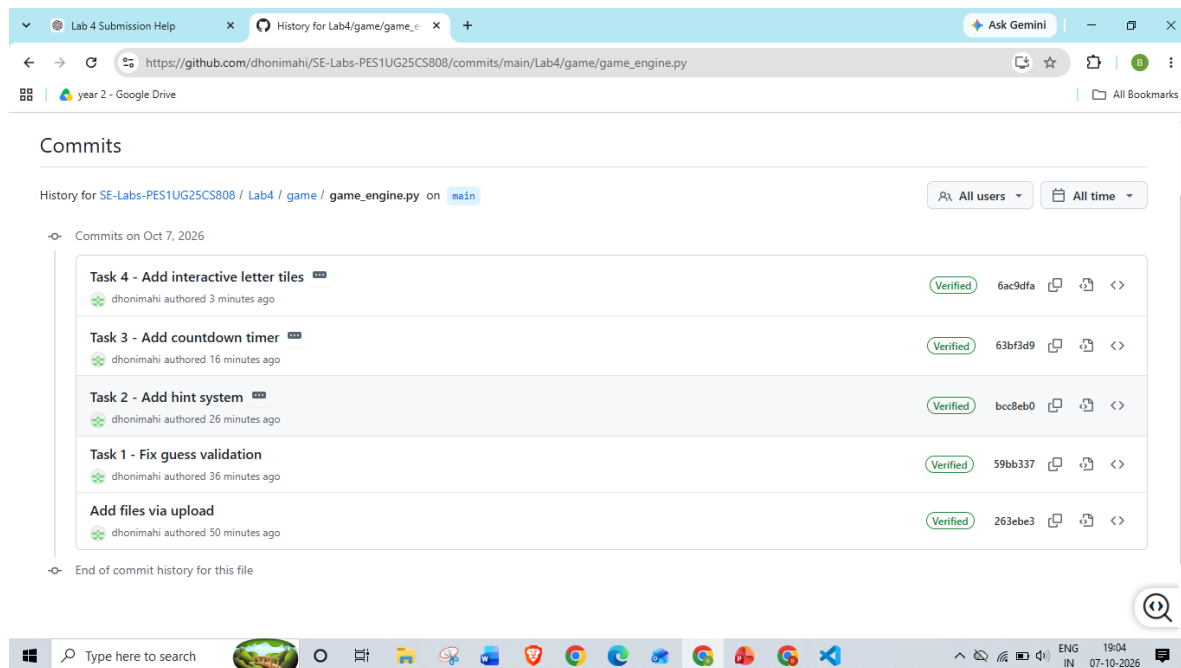
5. Prompt / Chat Evidence

The project was developed iteratively using an LLM as a debugging and pair-programming assistant. The public chat history contains the prompts and discussion used for the four Lab 4 tasks, including debugging, implementation, testing, and refinement.

Public Chat History: [Open public ChatGPT evidence](#)

- Fix the guess validation bug so the player's answer is compared with the actual secret word.
- Add a progressive hint system that reveals letters in correct positions and applies a score penalty.
- Add a countdown timer for each round and reveal the word when time runs out.
- Replace the static scrambled-word display with interactive graphical letter tiles.
- Make the hint feedback clearly state which letter and position were provided.

6. GitHub Commit Evidence



GitHub commit history showing separate commits for Tasks 1–4.

7. Final Result

The final Word Scramble Arena includes the four requested Lab 4 improvements: corrected guess validation, progressive hints with score penalties, a visible countdown timer with timeout handling, and interactive graphical letter tiles for rearranging the scrambled letters.